

Exp No: 01

Transformation between Cartesian, Cylindrical and spherical coordinate system

Aim:

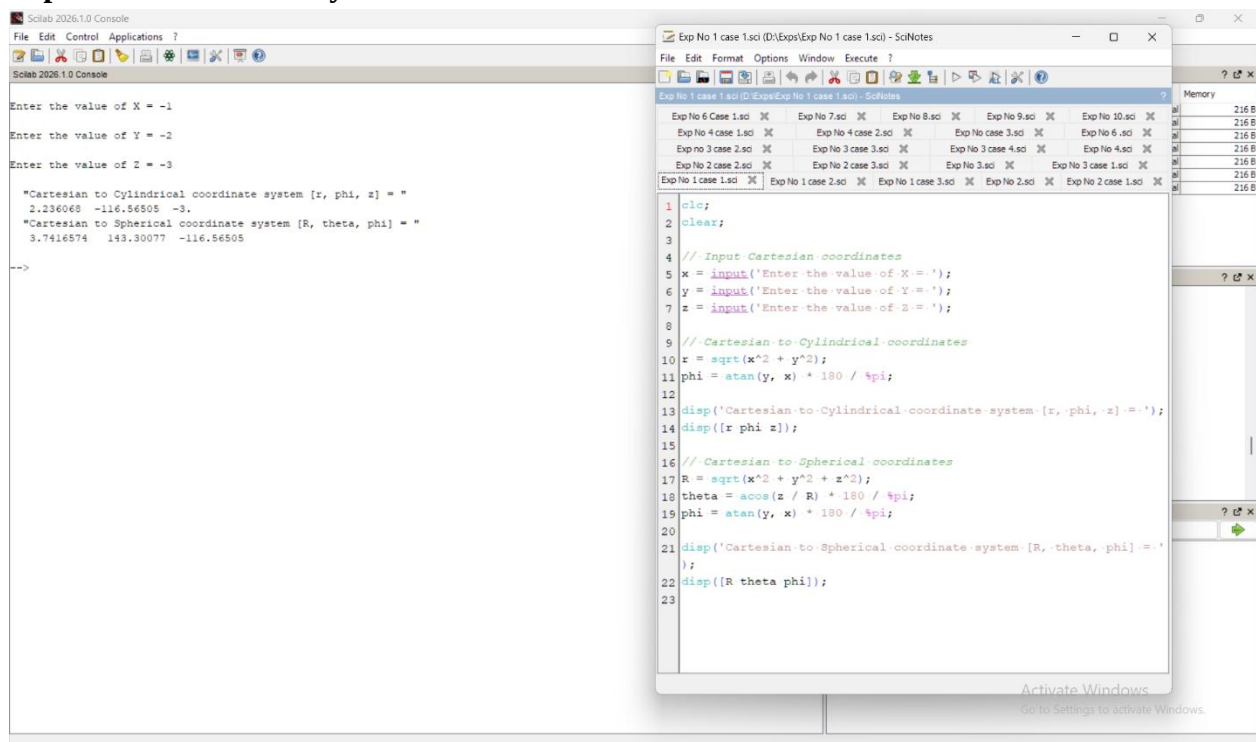
To write program for Transformation between Cartesian, Cylindrical and spherical coordinate system using SCILAB.

Software required:

- SCILAB version 6.0.1

Program:

Test case (i) Transformation of Cartesian coordinate points into Cylindrical and spherical coordinate system



The screenshot displays the SCILAB 2026.1.0 interface. On the left, the 'Console' window shows the execution of a script. It prompts for the values of X, Y, and Z, which are entered as -1, -2, and -3 respectively. The script then calculates the cylindrical coordinates (r, phi, z) and spherical coordinates (R, theta, phi). The results are displayed as follows:

```
Enter the value of X = -1
Enter the value of Y = -2
Enter the value of Z = -3

"Cartesian to Cylindrical coordinate system [r, phi, z] = "
2.236068 -116.56505 -3.
"Cartesian to Spherical coordinate system [R, theta, phi] = "
3.7416574 143.30077 -116.56505
```

On the right, the 'Scinotes' window shows the source code of the program:

```
1 clc;
2 clear;
3
4 // Input Cartesian coordinates
5 x = input('Enter the value of X = ');
6 y = input('Enter the value of Y = ');
7 z = input('Enter the value of Z = ');
8
9 // Cartesian to Cylindrical coordinates
10 r = sqrt(x^2 + y^2);
11 phi = atan(y, x) * 180 / %pi;
12
13 disp('Cartesian to Cylindrical coordinate system [r, phi, z] = ');
14 disp([r phi z]);
15
16 // Cartesian to Spherical coordinates
17 R = sqrt(x^2 + y^2 + z^2);
18 theta = acos(z / R) * 180 / %pi;
19 phi = atan(y, x) * 180 / %pi;
20
21 disp('Cartesian to Spherical coordinate system [R, theta, phi] = ');
22 disp([R theta phi]);
23
```

Practical Calculation:

Experiment observation

Exp 12

Test case (i) $\Rightarrow x = -1, y = -2, z = 3$

Cartesian $\rightarrow$ Cylindrical coordinates

$$r = \sqrt{x^2 + y^2} = \sqrt{(-1)^2 + (-2)^2} = \sqrt{1+4} = \sqrt{5}$$
$$r = 2.2361$$

Angle $\phi = \tan^{-1}(y/x) = \tan^{-1}\left(\frac{-2}{-1}\right)$ (III quadrant)

$x < 0$ & $y < 0$

$$\phi = 0.634349^\circ + 180^\circ$$
$$\phi = 243.435^\circ$$

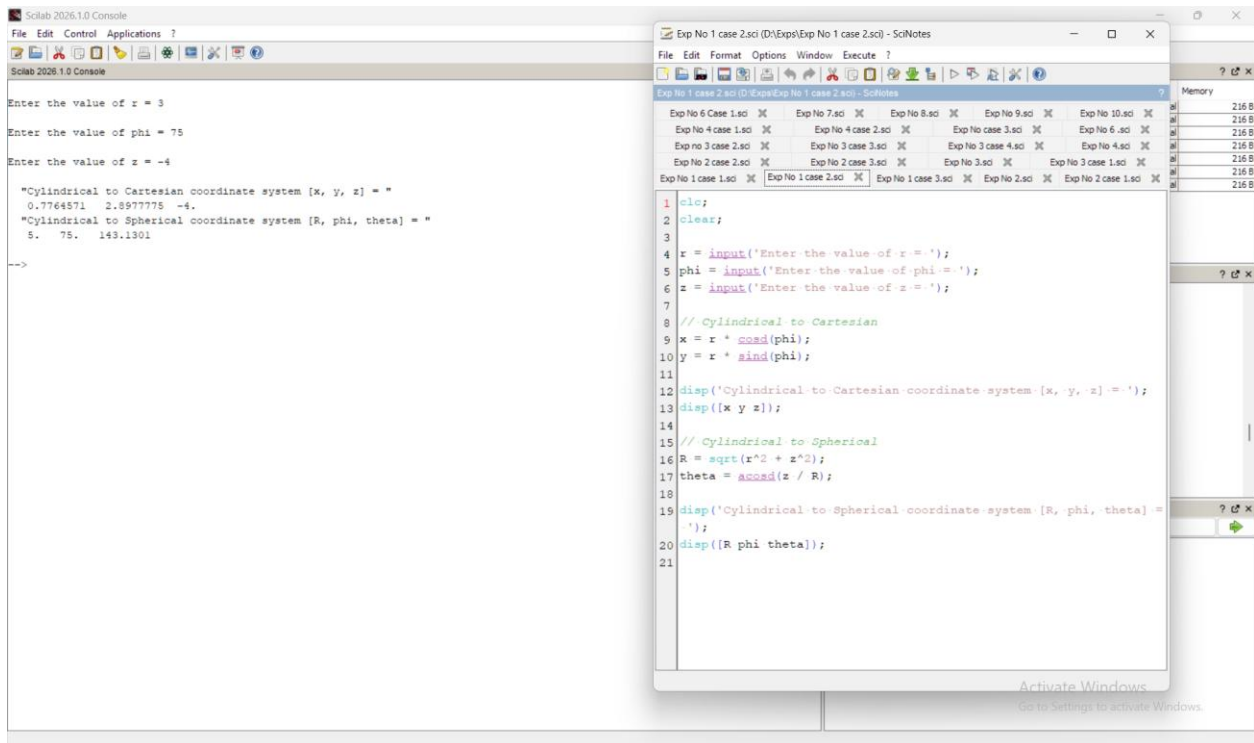
$z = -5$

ii) Cartesian $\rightarrow$ Cylindrical result

$$(r, \phi, z) = (2.2361, 243.435^\circ, -5)$$

Cartesian $\rightarrow$ Spherical coordinates

Test case (ii) Transformation of Cylindrical coordinate points into Cartesian and spherical coordinate system



Practical Calculation:

$$\begin{aligned}
 R &= \sqrt{x^2 + y^2 + z^2} \\
 &= \sqrt{(-1)^2 + (-2)^2 + (-3)^2} \\
 &= \sqrt{14} = 3.7417
 \end{aligned}$$

$\theta = \cos^{-1}\left(\frac{z}{R}\right)$
 $\theta = \cos^{-1}\left(\frac{-3}{3.7417}\right)$
 $\theta = \cos^{-1}(-0.8018)$
 $\theta = 143.301^\circ$

$\phi = \tan^{-1}(y/x)$
 since the point is in 3rd quadrant
 $\phi = 234.35^\circ$

Test case (iii) Transformation of spherical coordinate points into Cylindrical and Cartesian coordinate system

The screenshot shows a MATLAB environment with two windows. The 'Scilab 2026.1.0 Console' window on the left displays the execution of a script, showing prompts for input and the resulting coordinate transformations. The 'Exp No 1 case 3.sci' window on the right shows the source code of the script.

Scilab 2026.1.0 Console Output:

```
Enter the value of r = 1
Enter the value of phi = 150
Enter the value of theta = 80

*Spherical to Cartesian coordinate system [x, y, z] = "
-0.8528685  0.4924039  0.1736482
*Spherical to Cylindrical coordinate system [rho, phi, z] = "
0.9848078  150.  0.1736482
-->
```

Exp No 1 case 3.sci Script:

```
1 clc;
2 clear;
3
4 r = input('Enter the value of r = ');
5 phi = input('Enter the value of phi = ');
6 theta = input('Enter the value of theta = ');
7
8 // Spherical to Cartesian
9 x = r * sind(theta) * cosd(phi);
10 y = r * sind(theta) * sind(phi);
11 z = r * cosd(theta);
12
13 disp('Spherical to Cartesian coordinate system [x, y, z] = ');
14 disp([x y z]);
15
16 // Spherical to Cylindrical
17 rho = r * sind(theta);
18 z = r * cosd(theta);
19
20 disp('Spherical to Cylindrical coordinate system [rho, phi, z] = ');
21 disp([rho phi z]);
22
```

Practical Calculation:

$$\theta = \cos^{-1}(\tau/r) = \cos^{-1}(1/5) = 1.1071487 \text{ rad} \approx 63.43^\circ$$

$$(R, \phi, \theta) = (5, 75^\circ, 63.43^\circ)$$

Case (ii) $\theta = 80^\circ$ $\phi = 150^\circ$

$$r = 1$$

$$x = r \sin \theta \cos \phi$$

$$x = 1(\sin 80^\circ)(\cos 150^\circ)$$

$$= 1(0.9848)(-0.8660)$$

$$x = -0.8529$$

$$z = r \cos \theta$$

$$z = 1 \cos 80^\circ$$

$$z = 0.1736$$

$$y = r \sin \theta \sin \phi$$

$$y = 1(\sin 80^\circ)(\sin 150^\circ)$$

$$= 1(0.9848)(0.5)$$

$$y = 0.4924$$

$$(x, y, z) = (-0.8529, 0.4924, 0.1736)$$

Spherical to cylindrical $t = \theta$

$$\rho = r \sin \theta$$

$$\rho = 1 \sin 80^\circ$$

$$\rho = 0.9848, \phi = 150^\circ, z = 0.1736 //$$

Result:

Thus, the program was verified Transformation between Cartesian and Cylindrical coordinate system its classified two categories

- (a) Cartesian to cylindrical and spherical coordinate system
- (b) Cylindrical to cartesian and spherical coordinate system
- (c) spherical to cartesian and cylindrical coordinate system

using SCILAB was successfully.